

LI: Homework sheet 1 – Linear Algebra

Instructions. Please upload your solutions to the Canvas assignment page by 5pm, 5 November.

Plagiarism check: Your submission should be your own work and should not be identical or substantially similar to other submissions. A check for plagiarism will be performed on all submissions.

Assessment: This sheet is assessed, with a maximum contribution to your final mark of 5%.

Note. You should make sure that you state clearly any results you use in your proofs or derivations.

1. Let $(V, \langle \cdot, \cdot \rangle)$ denote a real inner product space and let $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ denote a basis for V .

Consider the Gram-Schmidt procedure applied to B : define $\mathbf{u}_1 := \mathbf{v}_1$ and compute

$$\mathbf{u}_{k+1} = \mathbf{v}_{k+1} - \sum_{j=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \mathbf{u}_j, \quad k = 1, 2, \dots, n-1.$$

- (a) Show by induction on k that $B_{k+1} := \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{k+1}\}$ is an orthogonal set for $k = 1, \dots, n-1$.
(b) What can you say about the resulting set B_n if the vectors $\mathbf{v}_1, \mathbf{v}_2$ in the basis B satisfy $\langle \mathbf{v}_1, \mathbf{v}_2 \rangle = 0$?
(c) Based on your answer for part (a), describe briefly what the output of the Gram-Schmidt algorithm will be in the case where the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ in B are mutually orthogonal, where $2 \leq m \leq n$.
2. Let $\mathcal{P}_n(\mathbb{R})$ denote the space of polynomials p of degree no greater than n and with real coefficients. When viewed as functions of a single real variable, say $p : [a, b] \rightarrow \mathbb{R}$, the polynomials in $\mathcal{P}_n(\mathbb{R})$ induce a vector space denoted by $\mathcal{P}_n([a, b])$:

$$\mathcal{P}_n([a, b]) := \{p(x) = a_0 + a_1x + \dots + a_nx^n \mid p : [a, b] \rightarrow \mathbb{R}, a_0, \dots, a_n \in \mathbb{R}\}.$$

This is an inner product space when equipped with the inner product $\langle \cdot, \cdot \rangle$, defined via

$$\langle p, q \rangle := \int_a^b p(x)q(x)dx.$$

Let $a = -1, b = 1$ and $U := \mathcal{P}_2([-1, 1])$. We denote by $\|\cdot\|$ the norm induced by $\langle \cdot, \cdot \rangle$ on U .

- (a) Check that the set $S = \{1, x\}$ is an orthogonal set in U with respect to $\langle \cdot, \cdot \rangle$. Find another *monic* polynomial¹ in U that is orthogonal to S .
(b) Let $q(x) = x^3$. Use part (a) to find $p^* \in U$ such that $\|q - p^*\| \leq \|q - p\|$ for all $p \in U$. Please state clearly any results you use in your derivation.

¹A polynomial of degree n is said to be monic if $a_n = 1$.